PROBLEM SET 3 DUE 3.29

1. Let

$$L = \mathbb{Z}\begin{bmatrix}1\\2\\3\end{bmatrix} + \mathbb{Z}\begin{bmatrix}4\\5\\6\end{bmatrix} + \mathbb{Z}\begin{bmatrix}7\\8\\9\end{bmatrix} \subseteq \mathbb{Z}^3$$

We have shown in class that

$$\mathbb{Z}^3/L \cong \mathbb{Z} \oplus \Big(\mathbb{Z}/3\mathbb{Z}\Big).$$

(1) Find a $\mathbb{Z}$-basis $\{\alpha_1, \alpha_2, \alpha_3\}$ of $\mathbb{Z}^3$ such that $\{\alpha_1, 3\alpha_2\}$ is a $\mathbb{Z}$-basis of L.

(2) Define the **saturation** of L to be

$$L' = \{\gamma \in \mathbb{Z}^3 : \exists n \geqslant 1 \text{ s.t. } n\gamma \in L\}.$$

Find a $\mathbb{Z}$-basis for L', and calculate the groups $\mathbb{Z}^3/L'$ and L'/L.

2. Let $f : M \to N$ and $f' : M' \to N'$ be A-module homomorphisms. Recall that $T(f, f')$ denotes the (unique) A-module homomorphism from $M \otimes_A M'$ to $N \otimes_A N'$ such that

$$T(f, f')(x \otimes x') = f(x) \otimes f'(x')$$

for all $x \in M$ and $x' \in M'$.

(1) Show that there exists a unique A-module homomorphism

$$\sigma : \mathrm{Hom}_A(M, N) \otimes_A \mathrm{Hom}_A(M', N') \to \mathrm{Hom}_A(M \otimes_A M', N \otimes_A N')$$

such that

$$\sigma(f \otimes f') = T(f, f')$$

for $f \in \mathrm{Hom}_A(M, N)$ and $f' \in \mathrm{Hom}_A(M', N')$.

(2) Let $g : N \to P$ and $g' : N' \to P'$ be A-module homomorphisms. Show that

$$T(gf, g'f') = T(g, g')T(f, f').$$

3. Let M, N, P be A-modules. Show that there exists a unique A-module homomorphism

$$\tau : \mathrm{Hom}_A(M, N) \otimes_A \mathrm{Hom}_A(N, P) \to \mathrm{Hom}_A(M, P)$$

such that $\tau(f \otimes g) = gf$ for $f \in \mathrm{Hom}_A(M, N)$ and $g \in \mathrm{Hom}_A(N, P)$.

4. Let A be a ring and $\mathfrak{a}, \mathfrak{b}$ be ideals of A. Show that

$$\Big(A/\mathfrak{a}\Big) \otimes_A \Big(A/\mathfrak{b}\Big) \cong A/(\mathfrak{a} + \mathfrak{b}).$$

5. Show that $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{Q}$.

6. Let $\Big\{M_i\Big\}_{i \in I}$ and N be A-modules, and let $\pi_i : \prod_{k \in I} M_k \to M_i$ be the i-th projection map. The family of A-module homomorphisms

$$\left\{T(\pi_i, \mathrm{id}_N) : \left(\prod_{k \in I} M_k\right) \otimes_A N \to M_i \otimes_A N\right\}_{i \in I}$$

induces an A-module homomorphism

$$\alpha : \left(\prod_{i\in I} M_i\right) \otimes N \to \prod_{i\in I} \Big(M_i \otimes N\Big);$$
$$(x_i)_i \otimes y \mapsto (x_i \otimes y)_i$$

(by the universal property of the direct product module on the RHS).

(1) Show that α is surjective if N is finitely generated.

(2) Show that

$$\alpha : \Big(\prod_{i=1}^{\infty} \mathbb{Z}\Big) \otimes_{\mathbb{Z}} \mathbb{Q} \to \prod_{i=1}^{\infty} \mathbb{Q}$$

is not surjective.